Algorithms

Dr. Mudassir Shabbir

LUMS University

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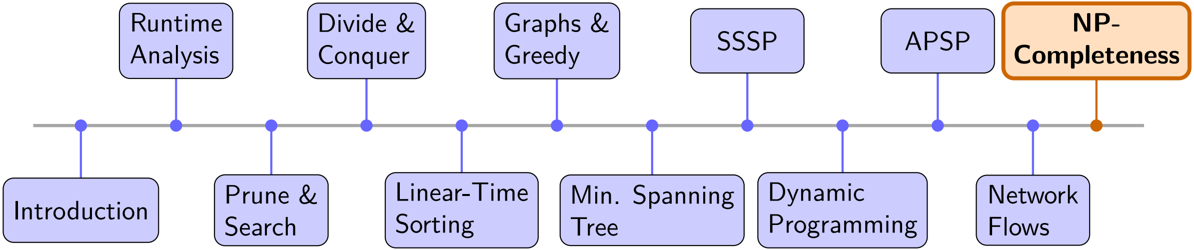
Recap & What's Next



NP-Completeness

A Catalogue of 29 NP-Complete Problems

Problems in This Catalogue

1. SAT
2. 3SAT
3. NAE-3SAT
4. Independent Set
5. Vertex Cover
6. Clique
7. Graph Coloring
8. Max Cut
9. Dominating Set
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12. Partition
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19. 3D-Matching
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21. Hamiltonian Path
22. Hamiltonian Cycle
23. Directed Ham. Path
24. Directed Ham. Cycle
25. Travelling Salesman (TSP)
26. Longest Path
27. Job Scheduling (Makespan)
28. Integer Linear Programming
29. Subgraph Isomorphism

1. Boolean Satisfiability (SAT)

Input

A Boolean formula $\varphi(x_1, \ldots, x_n)$ built from variables using $\wedge$, $\vee$, $\neg$.

Question

Does there exist a truth assignment $\tau \colon \{x_1, \ldots, x_n\} \to \{T, F\}$ such that $\varphi(\tau) = T$?

Significance. The first problem proved NP-complete (Cook 1971, Levin 1973). Every problem in NP reduces to SAT.

Standard form. SAT instances are usually given in *conjunctive normal form* (CNF): a conjunction of *clauses*, each clause a disjunction of *literals* (x_i or $\neg x_i$).

1. SAT — Example & Properties

Example.

$$\varphi = (\,x_1 \vee \neg x_2\,) \;\wedge\; (\,\neg x_1 \vee x_3 \vee x_2\,) \;\wedge\; \neg x_3$$

Assignment $x_1{=}T$, $x_2{=}F$, $x_3{=}F$: clause 1: T, clause 2: T, clause 3: T. ✓ Satisfiable.

Certificate (NP membership). The assignment τ. Evaluate φ bottom-up in $O(|\varphi|)$.

Special cases

- **2SAT** (each clause has ≤ 2 literals): in P via SCC on implication graph.
- **Horn-SAT** (each clause has ≤ 1 positive literal): in P via unit propagation.
- **XOR-SAT** (each clause is an XOR of literals): in P via Gaussian elimination.

2. 3-CNF Satisfiability (3SAT)

Input

A CNF formula φ in which every clause contains **exactly 3** literals (repetitions allowed, but all three from distinct variables).

Question

Does φ have a satisfying assignment?

Why 3SAT? The fixed clause width makes gadget constructions tractable. Most NP-hardness reductions start from 3SAT rather than general SAT.

Relation to SAT. 3SAT $\leq_p$ SAT trivially (it is a special case).
SAT $\leq_p$ 3SAT: replace each clause of length k by $O(k)$ clauses of length 3 using auxiliary variables.

2. 3SAT — Example & Properties

Example.

$$(x_1 \lor x_2 \lor x_3) \ \land \ (\neg x_1 \lor \neg x_2 \lor x_3) \ \land \ (x_1 \lor \neg x_2 \lor \neg x_3)$$

Assignment $x_1{=}T,\ x_2{=}F,\ x_3{=}T$: all three clauses satisfied. ✓

Unsatisfiable example.

$$(x \lor x \lor x) \ \land \ (\neg x \lor \neg x \lor \neg x)$$

Forces $x{=}T$ and $x{=}F$ simultaneously. ✗

Certificate. Assignment τ; check each clause has ≥ 1 true literal: $O(m)$.

Special cases

2SAT (2 literals per clause): in P.
MAX-2SAT (maximize satisfied clauses, 2 per clause): NP-hard.

3. Not-All-Equal 3SAT (NAE-3SAT)

Input

A 3-CNF formula φ (exactly 3 literals per clause).

Question

Does there exist an assignment such that in every clause, **at least one literal is true** *and* **at least one literal is false**?

(Equivalently: no clause is all-true or all-false.)

Contrast with 3SAT. 3SAT requires ≥ 1 true literal per clause. NAE-3SAT requires ≥ 1 true *and* ≥ 1 false literal per clause — a stricter condition.

Symmetry. If τ is a NAE-satisfying assignment, so is $\bar{\tau}$ (flip every variable). This makes NAE-3SAT useful in reductions that need a "symmetric" encoding.

3. NAE-3SAT — Example & Properties

Example. Clause $(x_1 \vee x_2 \vee x_3)$. NAE requires not all three to agree:

- $x_1{=}T, x_2{=}F, x_3{=}T$: true, false, true — NAE satisfied. ✓
- $x_1{=}T, x_2{=}T, x_3{=}T$: all true — NAE violated. ✗
- $x_1{=}F, x_2{=}F, x_3{=}F$: all false — NAE violated (also not a 3SAT solution). ✗

Certificate. Assignment τ; for each clause check $\exists$ true literal and $\exists$ false literal: $O(m)$.

Special cases

NAE-2SAT: in P (equivalent to 2-colourability, i.e. bipartiteness).
Monotone NAE-3SAT (no negated literals): still NP-complete.

4. Independent Set

Input

An undirected graph $G = (V, E)$ and an integer $k \geq 1$.

Question

Does G contain an **independent set** of size $\geq k$?

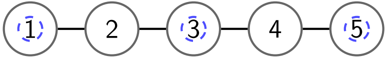
A set $S \subseteq V$ is *independent* if no two vertices in S are adjacent: $\forall u, v \in S : \ (u, v) \notin E$.

Intuition. Pick as many mutually non-conflicting elements as possible (scheduling, frequency assignment, etc.).

Key identity. S is an independent set in $G \iff S$ is a *clique* in the complement $\bar{G}$.

4. Independent Set — Example & Properties

Example. Path P_5:



$S = \{1, 3, 5\}$, size 3. No two are adjacent. ✓
Maximum IS for P_5 is 3.

Certificate. Set S; check $|S| \geq k$ and $\forall u, v \in S : (u, v) \notin E$: $O(k^2)$. ✓

Special cases

- **Bipartite graphs**: max IS $= n-$ max matching (König's theorem). In P via max flow.
- **Trees**: in P via DP.
- **Perfect graphs**: in P via ellipsoid method.

5. Vertex Cover

Input

An undirected graph $G = (V, E)$ and an integer $k \geq 0$.

Question

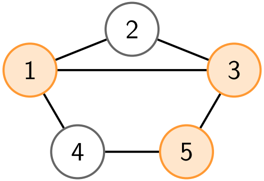
Does G have a **vertex cover** of size $\leq k$?

A set $C \subseteq V$ is a *vertex cover* if every edge has at least one endpoint in C:
$\forall (u, v) \in E : \; u \in C$ or $v \in C$.

Key identity. C is a vertex cover $\iff$ $V \setminus C$ is an independent set.

Consequence. G has a vertex cover of size k $\iff$ G has an independent set of size $|V| - k$.
So Vertex Cover and Independent Set are equivalent under polynomial reductions.

Example.



$C = \{1, 3, 5\}$ (orange): covers all 6 edges. ✓
Minimum VC has size 3.

Certificate. Set C; check $|C| \leq k$ and every edge covered: $O(|E|)$. ✓

Special cases

- **Trees**: in P via greedy leaf removal.
- **Bipartite graphs**: min VC $=$ max matching (König). In P.

Note. A 2-approximation exists: take both endpoints of any maximal matching.

6. Clique

Input

An undirected graph $G = (V, E)$ and an integer $k \geq 1$.

Question

Does G contain a **clique** of size $\geq k$?

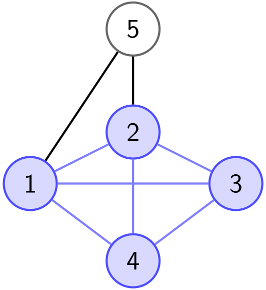
A set $S \subseteq V$ is a *clique* if every pair of vertices in S is adjacent: $\forall u \neq v \in S : \ (u, v) \in E$.

Key identity. S is a clique in $G \iff S$ is an independent set in the complement $\bar{G}$.

Intuition. Find the largest group of mutually-connected nodes (social network cliques, dense subgraphs, etc.).

6. Clique — Example & Properties

Example.



$\{1,2,3,4\}$ is a clique of size 4 (blue). ✓
Vertex 5 is not in the clique.

Certificate. Set S; verify all $\binom{k}{2}$ pairs are edges: $O(k^2)$. ✓

Special cases

- **Perfect graphs**: max clique in P (ellipsoid method).
- **Planar graphs**: max clique ≤ 4 by four-colour theorem; deciding clique-3 in P.

Inapproximability. No polynomial $n^{1-\varepsilon}$-approximation unless P $=$ NP.

7. Graph Coloring (k-Coloring)

Input

An undirected graph $G = (V, E)$ and an integer $k \geq 1$.

Question

Can the vertices of G be coloured with k colours such that no two adjacent vertices share a colour?

Formally: does there exist $f : V \to \{1, \ldots, k\}$ such that $\forall (u, v) \in E : \; f(u) \neq f(v)$?

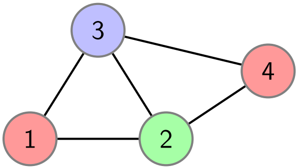
Chromatic number $\chi(G)$: smallest k for which G is k-colourable.

Decision version. Is $\chi(G) \leq k$?

Hardness threshold. k-Coloring is NP-complete for every fixed $k \geq 3$.

7. Graph Coloring — Example & Properties

Example.



$\chi(G) = 3$. Valid 3-colouring shown. ✓
2 colours are insufficient (C_3 subgraph).

Certificate. Colouring f; check every edge $O(|E|)$. ✓

Special cases

- $k = 1$: no edges required.
- $k = 2$: bipartiteness test via BFS. In P.
- **Perfect graphs**: $\chi(G) = \omega(G)$; in P.
- **Interval graphs**: in P by greedy sweep.

8. Maximum Cut (Max Cut)

Input

An undirected graph $G = (V, E)$ (possibly edge-weighted $w \colon E \to \mathbb{Z}^+$) and an integer k.

Question

Does there exist a partition $(S, V \setminus S)$ of V such that the total weight of edges crossing the cut is $\geq k$?

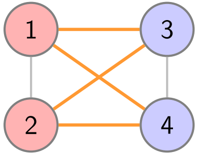
Unweighted: does $|\{(u, v) \in E : u \in S, v \notin S\}| \geq k$?

Contrast. *Minimum* s-t cut is in P (max-flow min-cut). Maximum cut is NP-complete.

Approximation. A random partition cuts $\frac{1}{2}$ of edges in expectation (2-approximation). The Goemans-Williamson SDP achieves a 0.878-approximation.

8. Max Cut — Example & Properties

Example. K_4:



$S = \{1, 2\}$, $V \setminus S = \{3, 4\}$. Cut size $= 4$. ✓ (Intra-partition edges (1-2) and (3-4) are not cut.)

Certificate. Partition $(S, V \setminus S)$; count crossing edges: $O(|E|)$. ✓

Special cases

- **Planar graphs**: max cut in P via T-join / matching.
- **Bipartite graphs**: max cut $= |E|$ (cut along the bipartition). In P.

9. Dominating Set

Input

An undirected graph $G = (V, E)$ and an integer $k \geq 1$.

Question

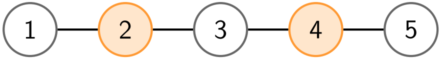
Does G have a **dominating set** of size $\leq k$?

A set $D \subseteq V$ is a *dominating set* if every vertex not in D is adjacent to at least one vertex in D: $\forall v \in V \setminus D : \exists u \in D$ s.t. $(u, v) \in E$.

Contrast with Vertex Cover. Vertex cover must "see" every *edge*; dominating set must "see" every *vertex*. Neither is a special case of the other in general.

9. Dominating Set — Example & Properties

Example. Path P_5:



$D = \{2, 4\}$ (orange): vertex 1 dominated by 2, vertex 3 dominated by 2 or 4, vertex 5 dominated by 4. ✓

Certificate. Set D; for each $v \notin D$ check $\exists$ neighbour in D: $O(|V| + |E|)$. ✓

Special cases

- **Trees**: in P via DP.
- **Interval graphs**: in P via greedy sweep.

Approximation. $\ln |V|$-approximation via greedy set cover.

10. Feedback Vertex Set (FVS)

Input

An undirected graph $G = (V, E)$ and an integer $k \geq 0$.

Question

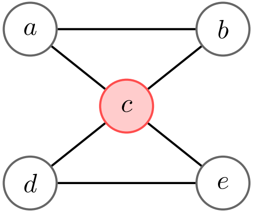
Does G have a **feedback vertex set** of size $\leq k$?

A set $F \subseteq V$ is a *feedback vertex set* if $G[V \setminus F]$ is acyclic (a forest): removing F eliminates all cycles.

Directed variant. Given a directed graph, does removing $\leq k$ vertices make it a DAG? Also NP-complete.

Related. Feedback *edge* set (remove $\leq k$ edges to make a forest) is also NP-complete.

Example.



Two triangles sharing vertex c.
$F = \{c\}$: removing c leaves $\{a, b\}$ and $\{d, e\}$ (two edges, no cycles). ✓

Certificate. Set F; verify $G[V \setminus F]$ is a forest using DFS: $O(|V| + |E|)$. ✓

Special cases

- **Tournaments** (directed): in P.
- $k = 0$: check acyclicity (DFS). In P.

FPT. Solvable in $O(4^k \cdot |V|)$ time (fixed-parameter tractable in k).

11. Subset Sum

Input

A multiset $S = \{a_1, a_2, \ldots, a_n\}$ of positive integers and a target integer $t > 0$.

Question

Does there exist a subset $I \subseteq \{1, \ldots, n\}$ such that $\sum_{i \in I} a_i = t$?

Why it's hard. The number of subsets is 2^n; no smarter-than-exponential exact algorithm is known in general.

Pseudo-polynomial algorithm. Dynamic programming solves it in $O(n \cdot t)$ time — polynomial in n and t, but exponential in $\log t$ (the *input size* of t). Hence NP-complete but "weakly" so.

11. Subset Sum — Example & Properties

Example. $S = \{3, 7, 1, 8, 5, 2\}$, target $t = 14$.
Try $\{7, 5, 2\} = 14$. ✓ Try $\{8, 5, 1\} = 14$. ✓ (Multiple solutions exist.)

Infeasible example. $S = \{2, 4, 6\}$, target $t = 5$. All subsets give even sums; 5 is odd. ✗

Certificate. Subset I; sum $|I|$ numbers and compare to t: $O(n)$. ✓

Special cases

- $t = O(n^c)$ for constant c: pseudo-polynomial DP in polynomial time.
- All a_i are given in unary: in P (DP runs in polynomial time in unary input size).

12. Partition

Input

A multiset $S = \{a_1, a_2, \ldots, a_n\}$ of positive integers.

Question

Can S be partitioned into two subsets S_1 and $S_2 = S \setminus S_1$ such that $\sum_{a \in S_1} a = \sum_{a \in S_2} a$?

Reduction from Subset Sum. Partition $\leq_p$ Subset Sum: the two halves must each sum to $\frac{T}{2}$ where $T = \sum a_i$.

Reduction to Subset Sum. Partition $\leq_p$ Subset Sum with $t = T/2$.

Consequence. Partition and Subset Sum are equivalent under polynomial reductions.

12. Partition — Example & Properties

Example. $S = \{3, 1, 1, 2, 2, 1\}$. Total $T = 10$. Need two halves summing to 5.
$S_1 = \{3, 2\}$: sum 5. $S_2 = \{1, 1, 2, 1\}$: sum 5. ✓

Infeasible example. $S = \{1, 2, 3, 5\}$. Total $T = 11$ (odd) — no equal partition possible. ✗

Certificate. Partition (S_1, S_2); sum both sides: $O(n)$. ✓

Special cases

- T is small (pseudo-poly DP in $O(nT)$).
- Identical elements: trivially feasible iff n is even.

Note. The $\frac{4}{3}$-approximation for Makespan uses Partition as a subroutine reduction.

13. 0-1 Knapsack

Input

n items, each with weight $w_i > 0$ and value $v_i > 0$; knapsack capacity $W > 0$; value target V.

Question

Does there exist a subset $I \subseteq \{1, \dots, n\}$ such that

$$\sum_{i \in I} w_i \leq W \quad \text{and} \quad \sum_{i \in I} v_i \geq V\,?$$

"0-1": each item is taken entirely or not at all (no fractions).

Reduction from Subset Sum. Set $w_i = v_i = a_i$, $W = V = t$. An exact-weight subset a feasible knapsack meeting both constraints.

13. 0-1 Knapsack — Example & Properties

Example. $W = 10$, $V = 15$.

Item	Weight	Value	Selected?
1	6	10	✓
2	4	8	✓
3	3	5	✗
4	5	7	✗

Total weight $= 10 \leq W$, total value $= 18 \geq 15$. ✓

Certificate. Subset I; sum weights and values: $O(n)$. ✓

Special cases

- **Fractional Knapsack** (fractions allowed): in P via greedy (sort by v_i/w_i).
- **Pseudo-polynomial DP**: $O(nW)$ — polynomial if W is small.

14. Bin Packing

Input

n items with sizes $s_1, \ldots, s_n \in (0, 1]$; bin capacity 1; integer k.

Question

Can all n items be packed into $\leq k$ bins such that the total size in each bin is ≤ 1?

Reduction from Partition. Given $\{a_1, \ldots, a_n\}$ with $\sum a_i = 2T$, set item sizes $s_i = a_i/T$. Two bins suffice iff the items partition into two halves of total size 1 each.

Practical context. Packing files onto disks, scheduling jobs onto machines, cutting stock problems.

14. Bin Packing — Example & Properties

Example. Items of sizes $0.5,\ 0.7,\ 0.3,\ 0.6,\ 0.4$. Can we use $k = 2$ bins?
Bin 1: $\{0.7, 0.3\} = 1.0$. Bin 2: $\{0.5, 0.4\} = 0.9$. Item 0.6 needs a third bin. ✗
With $k = 3$: Bin 1: $\{0.7, 0.3\}$, Bin 2: $\{0.6, 0.4\}$, Bin 3: $\{0.5\}$. ✓

Certificate. Assignment of items to $\leq k$ bins; check each bin's total: $O(n)$. ✓

Special cases

- **All items size** $1/k$: trivially pack k per bin. In P.
- **Fixed** k: DP in $O(n^k)$; polynomial for fixed k.

Approx. First-Fit Decreasing gives a $\frac{11}{9}\text{OPT} + \frac{6}{9}$ guarantee.

15. Set Cover

Input

A universe $U = \{u_1, \ldots, u_m\}$, a collection $\mathcal{S} = \{S_1, \ldots, S_n\}$ of subsets of U, and an integer k.

Question

Does there exist a subcollection $\mathcal{C} \subseteq \mathcal{S}$ with $|\mathcal{C}| \leq k$ such that $\bigcup_{S \in \mathcal{C}} S = U$?

Reduction from Vertex Cover. Given $G = (V, E)$: set $U = E$, and for each vertex $v \in V$ create $S_v = \{e \in E : v \in e\}$. A vertex cover of size $k \iff$ a set cover of size k.

Greedy approximation. Repeatedly pick the set covering the most uncovered elements: $\ln |U|$-approximation. Optimal unless P = NP (Feige 1998).

15. Set Cover — Example & Properties

Example. $U = \{1,2,3,4,5\}$, $\mathcal{S} = \{S_1, S_2, S_3, S_4\}$:

$$S_1 = \{1,2,3\},\ S_2 = \{2,4\},\ S_3 = \{3,4\},\ S_4 = \{4,5\}.$$

$k = 2$: $\{S_1, S_4\} = \{1,2,3\} \cup \{4,5\} = U$. ✓

Certificate. Subcollection $\mathcal{C}$; take union and compare to U: $O(k \cdot |U|)$. ✓

Special cases

- $k = 1$: check if any single set equals U. In P.
- **All sets have size** ≤ 2: bipartite matching / 2-SAT. In P.
- **Sets form intervals on a line**: polynomial greedy.

16. Set Packing

Input

A universe U, a collection $\mathcal{S} = \{S_1, \ldots, S_n\}$ of subsets of U, and an integer k.

Question

Does there exist a subcollection $\mathcal{P} \subseteq \mathcal{S}$ with $|\mathcal{P}| \geq k$ such that all sets in $\mathcal{P}$ are **pairwise disjoint**?

Contrast with Set Cover. Set Cover: cover everything using few sets. Set Packing: find many sets that don't overlap.

Reduction from Independent Set. Given $G = (V, E)$: set $U = E$, and for each vertex v let $S_v = \{e \in E : v \in e\}$. Two sets S_u, S_v are disjoint iff $(u, v) \notin E$, so a packing of size k IS of size k.

16. Set Packing — Example & Properties

Example. $U = \{1, \ldots, 6\}$, sets $\{1, 2, 3\}$, $\{3, 4, 5\}$, $\{4, 5, 6\}$, $\{1, 6\}$.
$k = 2$: $\{\{1, 2, 3\}, \{4, 5, 6\}\}$ — disjoint. ✓
$k = 3$: no 3 pairwise disjoint sets from this collection (each shares at least one element). ✗

Certificate. Collection $\mathcal{P}$; verify pairwise disjointness: $O(k^2 \cdot |U|)$. ✓

Special cases

- **All sets have size** ≤ 2: maximum matching. In P.
- **Sets form intervals**: interval scheduling, greedy. In P.

17. Exact Cover

Input

A universe $U = \{u_1, \ldots, u_m\}$ and a collection $\mathcal{S} = \{S_1, \ldots, S_n\}$ of subsets of U (subsets may have any size).

Question

Does there exist a subcollection $\mathcal{C} \subseteq \mathcal{S}$ such that every element of U appears in **exactly one** set in $\mathcal{C}$?

Equivalently: the sets in $\mathcal{C}$ are pairwise disjoint and $\bigcup_{S \in \mathcal{C}} S = U$.

Position in the hierarchy.

- **Set Cover** $\supseteq$ Exact Cover: Set Cover only requires $\bigcup S = U$; Exact Cover additionally forbids overlap.
- **Exact Cover** $\supseteq$ X3C: X3C is Exact Cover restricted to $|U| = 3q$ and all sets of size exactly 3.

17. Exact Cover — Example & Properties

Example. $U = \{1, 2, 3, 4, 5, 6\}$, $\mathcal{S} = \{\{1, 2\},\ \{3, 4, 5\},\ \{6\},\ \{2, 3\},\ \{4, 5, 6\}\}$.
$\mathcal{C} = \{\{1, 2\}, \{3, 4, 5\}, \{6\}\}$: disjoint, union $= U$. ✓
$\mathcal{C}' = \{\{1, 2\}, \{4, 5, 6\}, \{3, 4\}\}$: element 4 appears twice. ✗

Certificate. Subcollection $\mathcal{C}$; verify pairwise disjointness and $\bigcup \mathcal{C} = U$: $O(|\mathcal{C}| \cdot |U|)$. ✓

Special cases

- **All sets disjoint**: trivially check if they cover U. In P.
- **Sets of size** ≤ 2: bipartite matching / 2-SAT reduction. In P.
- **Interval sets on a line**: polynomial greedy (no overlaps needed).

18. Exact Cover by 3-Sets (X3C)

Input

A universe U with $|U| = 3q$ and a collection $\mathcal{C}$ of 3-element subsets of U.

Question

Does there exist a subcollection $M \subseteq \mathcal{C}$ of exactly q sets such that every element of U appears in **exactly one** set in M?

Relation to Set Cover. X3C is a special case of Exact Cover (all sets have size 3, target is $|U|/3$ sets). General Exact Cover $\leq_p$ X3C (X3C captures the essence of covering with equal-size sets).

Use in reductions. X3C serves as an intermediate step in proving 3DM NP-hard via 3SAT $\leq_p$ X3C $\leq_p$ 3DM.

18. X3C — Example & Properties

Example. $U = \{1,2,3,4,5,6\}$, $q = 2$.
$\mathcal{C} = \{\{1,2,3\},\ \{4,5,6\},\ \{1,4,5\},\ \{2,3,6\}\}$.
$M = \{\{1,2,3\},\{4,5,6\}\}$: 2 sets, each element appears exactly once. ✓

$M' = \{\{1,4,5\},\{2,3,6\}\}$: also valid — another exact cover. ✓

Certificate. Subcollection M; verify $|M| = q$ and pairwise disjointness: $O(q^2)$. ✓

Special cases

- **All sets disjoint in the collection**: exact cover is straightforward. In P.
- $|U| = 3$ (single triple): trivial.

19. 3-Dimensional Matching (3DM)

Input

Disjoint sets B, G, P each of size n, and a set $T \subseteq B \times G \times P$ of triples.

Question

Does there exist a **perfect matching** $M \subseteq T$ — a set of n pairwise disjoint triples that together cover every element of $B \cup G \cup P$?

Relation to matching. 2-dimensional matching (bipartite perfect matching) is in P via max-flow. 3DM with $|B| = |G| = |P| = n$ requires matching across three sets simultaneously — NP-complete.

Special case of X3C. 3DM is X3C where $U = B \cup G \cup P$ and every set contains one element from each part.

19. 3DM — Example & Properties

Example. $B = \{b_1, b_2\}$, $G = \{g_1, g_2\}$, $P = \{p_1, p_2\}$.
$T = \{(b_1, g_1, p_1),\ (b_1, g_2, p_2),\ (b_2, g_1, p_2),\ (b_2, g_2, p_1)\}$.
$M = \{(b_1, g_1, p_1),\ (b_2, g_2, p_2)\}$: 2 disjoint triples, all 6 elements covered. ✓
$M' = \{(b_1, g_1, p_1),\ (b_2, g_2, p_1)\}$: p_1 appears twice. ✗

Certificate. M; verify $|M| = n$ and pairwise disjointness: $O(n^2)$. ✓

Special cases

- **2D matching**: bipartite perfect matching. In P (max-flow).
- k-**DM for** $k \geq 3$: NP-complete by the same reduction chain.

20. Steiner Tree

Input

An undirected graph $G = (V, E)$ with edge weights $w \colon E \to \mathbb{Z}^+$, a set of *terminals* $R \subseteq V$, and an integer k.

Question

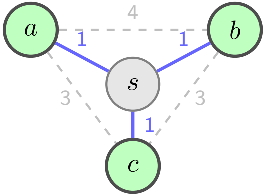
Does there exist a subtree of G that spans all terminals in R and has total edge weight $\leq k$?

Steiner points. The optimal tree may include non-terminal "Steiner" vertices from $V \setminus R$ to reduce total weight.

Contrast. If $R = V$: minimum spanning tree (in P). If $|R| = 2$: shortest path (in P). General Steiner tree: NP-complete.

20. Steiner Tree — Example & Properties

Example. Terminals $R = \{a, b, c\}$ (bold), Steiner point s:



Steiner tree via s: cost 3. Direct spanning tree (no s): cost ≥ 6.

Certificate. The subtree; verify it spans R and sum weights $\leq k$: $O(|V|)$. ✓

Special cases

- $|R| = 2$: shortest path. In P.
- $R = V$: MST. In P.
- **Series-parallel graphs**: in P.

Approx. $2(1 - 1/|R|)$-approximation (MST on metric closure of R).

21. Hamiltonian Path

Input

An undirected graph $G = (V, E)$.

Question

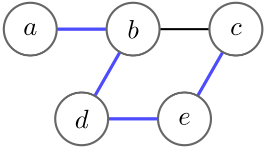
Does G contain a **Hamiltonian path** — a simple path that visits every vertex exactly once?

Variants.

- s-t **Ham-Path**: does a Ham. path from s to t exist?
- **Existence**: does *any* Ham. path exist? Both variants are NP-complete.

Contrast. Finding *any* path between two vertices is in P (BFS/DFS). Requiring the path to visit *all* vertices is NP-complete — a subtle but crucial difference.

Example.



Ham. path: a–b–d–e–c (blue). ✓

Certificate. Vertex sequence; verify each consecutive pair is an edge, all distinct: $O(|V|)$. ✓

Special cases

- **Tournament graphs** (complete directed): always has Ham. path. In P.
- **Dirac's condition**: $\delta(G) \geq |V|/2$ guarantees Ham. cycle. But deciding given graph has Ham. path is NP-hard.

22. Hamiltonian Cycle

Input

An undirected graph $G = (V, E)$.

Question

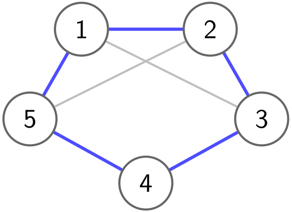
Does G contain a **Hamiltonian cycle** — a simple cycle that visits every vertex exactly once?

Relation to Ham-Path. Ham-Path $\leq_p$ Ham-Cycle: add a new vertex w connected to all vertices of G; then G' has a Ham. cycle iff G has a Ham. path.

Relation to TSP. Ham-Cycle $\leq_p$ TSP: set edge weights 0 for existing edges, 1 for missing; ask for a tour of cost 0.

22. Hamiltonian Cycle — Example & Properties

Example.



Ham. cycle: $1\text{—}2\text{—}3\text{—}4\text{—}5\text{—}1$ (blue). ✓

Certificate. Vertex permutation forming a cycle; verify edges and coverage: $O(|V|)$. ✓

Special cases

- **Complete graphs** K_n: trivially has a Ham. cycle. In P.
- $K_{n,n}$ **(bipartite)**: Ham. cycle exists for $n \geq 2$. In P.
- **Regular planar graphs**: NP-complete in general.

23. Directed Hamiltonian Path

Input

A directed graph $G = (V, E)$ where each edge has a direction.

Question

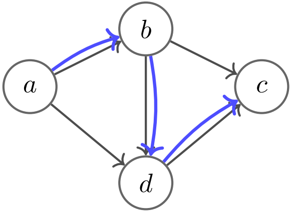
Does G contain a **directed Hamiltonian path** — a directed simple path that visits every vertex exactly once, following edge directions?

Relation to undirected Ham-Path. Undirected Ham-Path $\leq_p$ Directed Ham-Path: replace each undirected edge $\{u, v\}$ with a *gadget* (split each vertex into in/out copies to enforce direction constraints).

Acyclic case. In a DAG, a Ham. path exists iff the DAG has a unique topological sort — decidable in $O(|V| + |E|)$. DAGs are the key poly special case.

23. Directed Hamiltonian Path — Example & Properties

Example.



Directed Ham. path: $a \to b \to d \to c$ (blue). ✓

Certificate. Vertex sequence; verify each consecutive edge exists in the right direction: $O(|V|)$. ✓

Special cases

- **DAGs**: unique topological sort Ham. path. In P.
- **Tournaments**: always has a directed Ham. path. In P.

24. Directed Hamiltonian Cycle

Input

A directed graph $G = (V, E)$.

Question

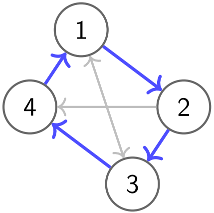
Does G contain a **directed Hamiltonian cycle** — a directed simple cycle that visits every vertex exactly once?

Relation to directed Ham-Path. Directed Ham-Path $\leq_p$ Directed Ham-Cycle: add a vertex w with edges from every vertex to w and from w to every vertex.

Relation to TSP. The directed Ham-Cycle reduction to TSP gives a directed TSP formulation — all edge weights 0 for existing arcs, 1 for missing arcs; budget 0.

24. Directed Hamiltonian Cycle — Example & Properties

Example.



Directed Ham. cycle: $1 \to 2 \to 3 \to 4 \to 1$. ✓

Certificate. Vertex permutation; verify directed edges form a cycle covering all vertices: $O(|V|)$. ✓

Special cases

- **Strongly connected tournaments**: always has a directed Ham. cycle. In P.
- **Strongly connected planar digraphs**: NP-complete in general.

25. Travelling Salesman Problem (TSP)

Input

A complete weighted graph on n cities (vertices) with pairwise distances $d_{ij} \geq 0$ and a budget B.

Question

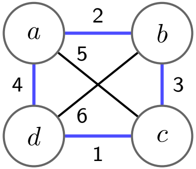
Does there exist a **tour** — a Hamiltonian cycle — of total weight $\leq B$?

Reduction from Ham-Cycle. Given $G = (V, E)$: set $d_{uv} = 0$ if $(u, v) \in E$, else $d_{uv} = 1$. Budget $B = 0$. A tour of cost 0 uses only original edges G has a Ham. cycle.

Approximation.

- **Metric TSP** (triangle inequality): Christofides gives a 3/2-approximation.
- **General TSP**: no constant-factor approximation unless P = NP.

Example. 4 cities with distances:



Tour a–b–c–d–a: cost $2+3+1+4=10$. Budget $B=10$: ✓

Certificate. Tour (permutation of cities); sum n edge weights: $O(n)$. ✓

Special cases

n **cities on a line** (sorted): optimal tour is just back-and-forth. In P.

26. Longest Path

Input

An undirected graph $G = (V, E)$ (possibly edge-weighted $w\colon E \to \mathbb{Z}^+$) and an integer k.

Question

Does G contain a **simple path** of length $\geq k$?

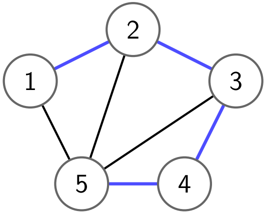
Unweighted: a simple path using $\geq k$ edges.
Weighted: a simple path with total weight $\geq k$.

Reduction from Ham-Path. Set $k = |V| - 1$. A simple path of length $|V| - 1$ visits exactly $|V|$ vertices — that is a Hamiltonian path. The reduction is the identity (same graph, target $k = |V| - 1$).

Contrast. Shortest path is in P (Dijkstra, BFS). Longest path is NP-complete. The two problems are *not* symmetric.

26. Longest Path — Example & Properties

Example. Graph on 5 vertices:



Simple path $1{-}2{-}3{-}4{-}5$: length 4. For $k = 4$: ✓

This is also a Hamiltonian path.

Certificate. Vertex sequence; verify edges and simplicity, count length: $O(|V|)$. ✓

Special cases

- **DAGs**: longest path in P via DP in topological order.
- **Trees**: longest path (diameter) in P via two BFS.

27. Job Scheduling — Makespan Minimisation

Input

n jobs with processing times $p_1, \ldots, p_n > 0$; m identical parallel machines; target makespan T.

Question

Can all n jobs be assigned to the m machines such that every machine finishes by time T?

Equivalently: does there exist a partition of jobs into m groups such that the maximum group sum is $\leq T$?

Special case ($m = 2$). Two-machine scheduling $\equiv$ Partition: can jobs be split into two groups with equal total processing time? NP-complete by reduction from Partition.

Denoted $P||C_{\max}$ in scheduling notation (identical machines, no constraints, minimize makespan).

27. Job Scheduling — Example & Properties

Example. $m = 2$ machines, jobs with times $\{3, 4, 2, 5, 1\}$, target $T = 8$.
Machine 1: jobs $\{4, 3, 1\}$ — total 8. ✓
Machine 2: jobs $\{5, 2\}$ — total 7 ≤ 8. ✓

Certificate. Assignment of jobs to machines; sum each machine's load: $O(n)$. ✓

Special cases

- $m = 1$: trivially feasible iff $\sum p_i \leq T$.
- **All** $p_i = 1$: assign greedily round-robin. In P.
- **Fixed** m: DP in pseudo-polynomial $O(n \cdot T^m)$ (poly for fixed m, small T).

Approx. List scheduling (LPT): $(4/3 - 1/3m)$-approximation.

28. Integer Linear Programming (ILP)

Input

A matrix $A \in \mathbb{Z}^{m \times n}$, vectors $b \in \mathbb{Z}^m$, $c \in \mathbb{Z}^n$, and an integer k.

Question

Does there exist an integer vector $x \in \mathbb{Z}^n$ such that

$$Ax \leq b \quad \text{and} \quad c^\top x \geq k\,?$$

Contrast. Linear Programming (LP) — same problem over $\mathbb{R}^n$ — is in P (ellipsoid method, interior point). Requiring $x \in \mathbb{Z}^n$ makes it NP-complete.

Generality. Many NP-complete problems (Knapsack, Subset Sum, Vertex Cover, etc.) can be modelled as ILPs. ILP is therefore a very general NP-complete problem.

Example.

$$\text{maximize } 3x_1 + 5x_2 \quad \text{subject to} \quad x_1 + x_2 \le 4,\ x_1, x_2 \ge 0,\ x_1, x_2 \in \mathbb{Z}.$$

LP optimum: $x_1 = 0,\ x_2 = 4$ (value 20). Integer optimum: same here — $(0,4)$ is already integer. ✓
More interesting example: $x_1 + 2x_2 \le 5$, maximize $x_1 + x_2$, integers. LP gives $x_2 = 2.5$ (non-integer); ILP gives $x_2 = 2, x_1 = 1$ (value 3).

Certificate. Integer vector x; check $Ax \le b$ and $c^\top x \ge k$: $O(mn)$. ✓

Special cases

- **LP** (real-valued): in P via ellipsoid / interior point.
- **0-1 ILP on totally unimodular matrices**: LP relaxation is always integral. In P.
- **Fixed number of variables** n: in P (Lenstra's theorem).

29. Subgraph Isomorphism

Input

Two undirected graphs $G = (V_G, E_G)$ and $H = (V_H, E_H)$.

Question

Does G contain a subgraph isomorphic to H? That is, does there exist an injective mapping $f : V_H \to V_G$ such that for every edge $(u, v) \in E_H$, the pair $(f(u), f(v))$ is an edge in G?

Relation to other problems.

- **Clique**: H is a complete graph K_k; asks if G has a k-clique. NP-complete.
- **Hamiltonian Path**: H is a path on $|V_G|$ vertices; asks if G has a Ham. path. NP-complete.

Generalisation. Subgraph Isomorphism generalises many NP-complete problems, making it a very powerful NP-complete problem.